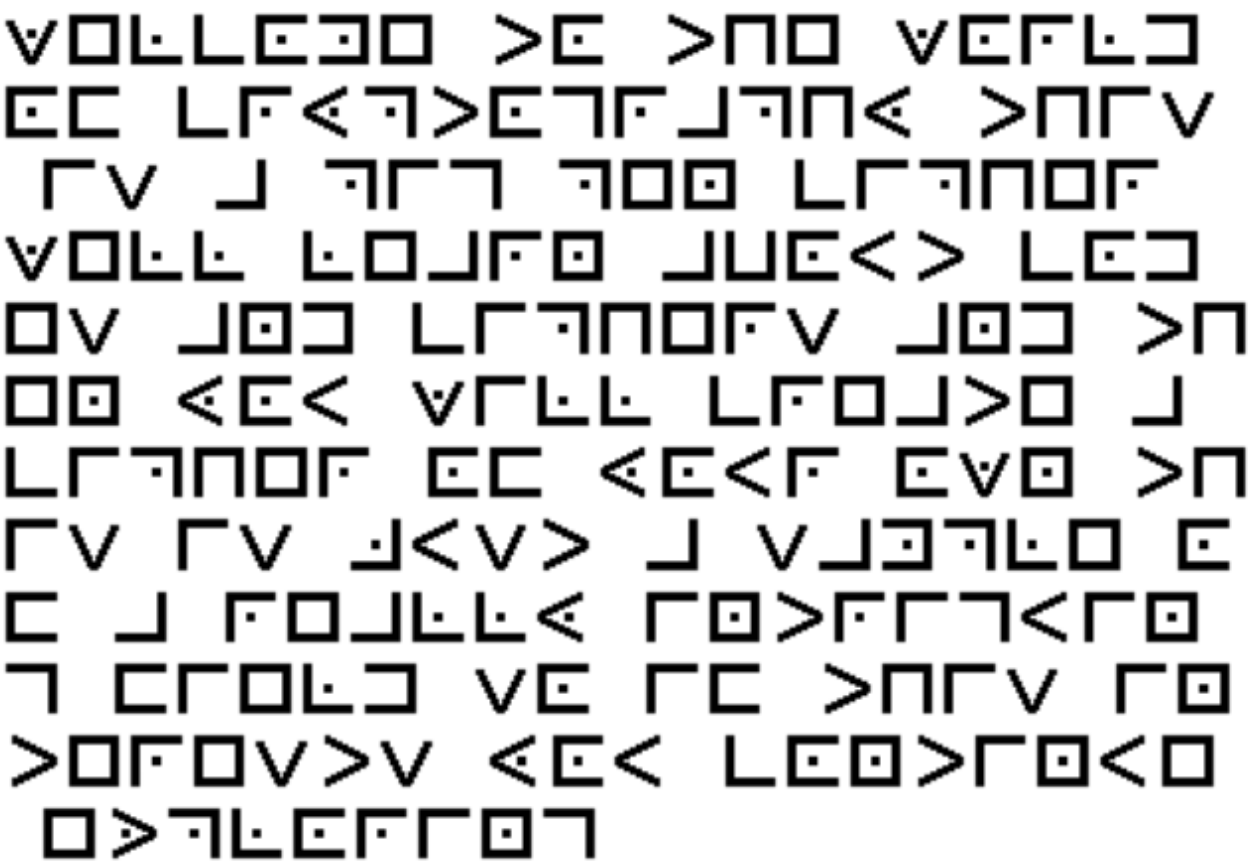


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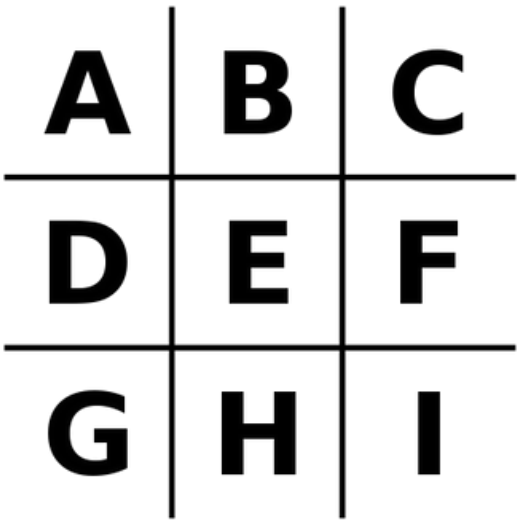
WELCOME

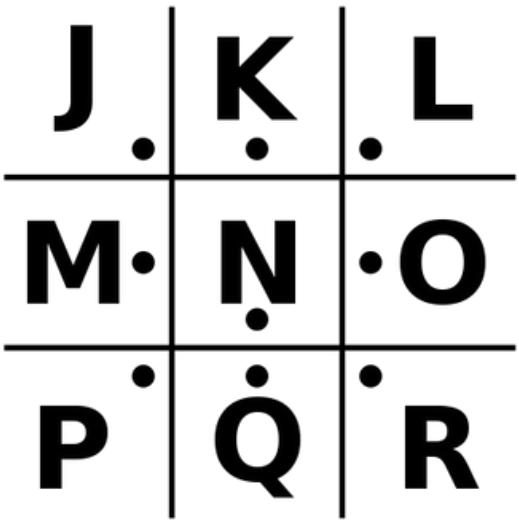


- Cryptography (Wikipedia)
- Cryptanalysis (Wikipedia)
- Cipher (Wikipedia)
- Code (Wikipedia)

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THE KEY









Source: Wikipedia

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CODES

A code involves replacing words or phrases with symbols or unrelated terms, typically necessitating the use of a codebook to decipher. Examples:

The eagle has landed.

The quick brown 狐 jumps 上 the lazy 犬

Telegraph Code
INSANE = at what price, free on board and freight,
can you offer us cotton for shipment
by steamer sailing this week?

Develop codes that you could use with your friends:

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CIPHERS

Ciphers use algorithms to replace letters, groups of letters, bits, or blocks of bits. **Use one of the ciphers below to encrypt your name, and for added practice, encrypt this Walt Disney quotation:**

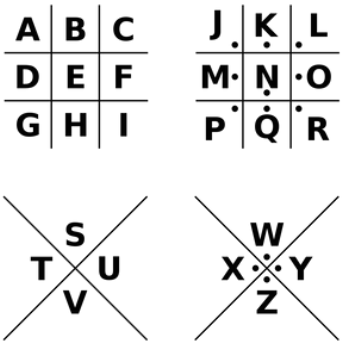
I f y o u c a n d r e a m i t ,

y o u c a n d o i t .

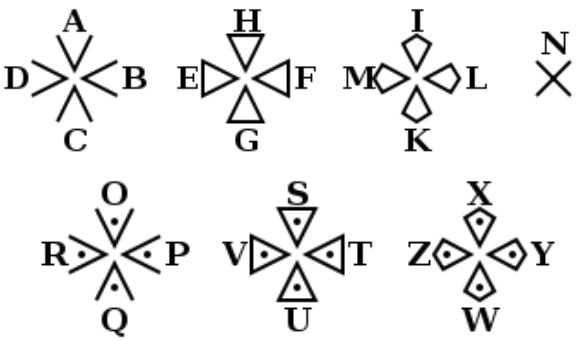
A Caesar Cipher

Plain	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Cipher	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W

The Pigpen Cipher



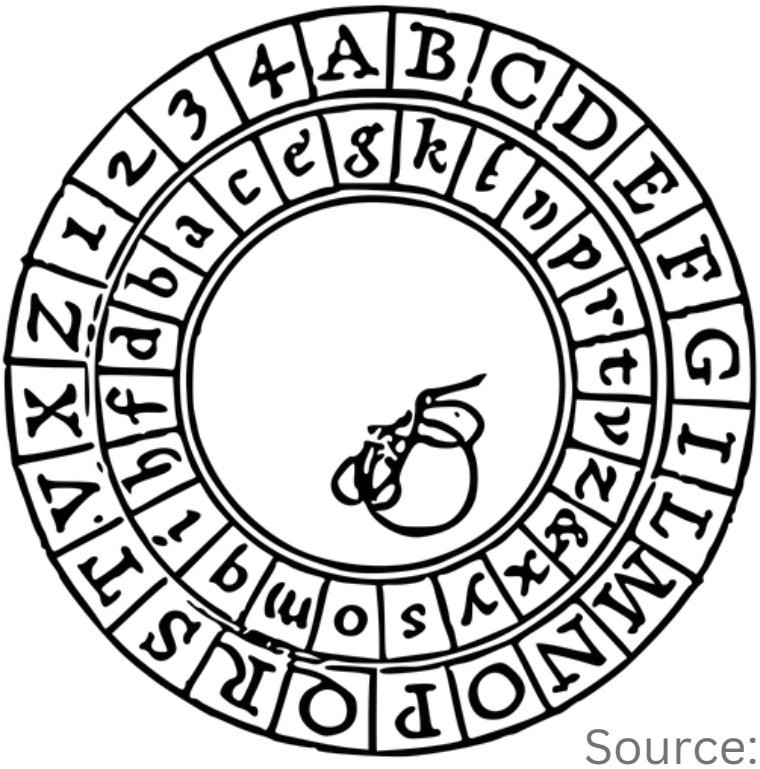
The Templar Cipher



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MAKE IT

In 1467, architect Leon Alberti created the Alberti cipher. **Make one for yourself.**



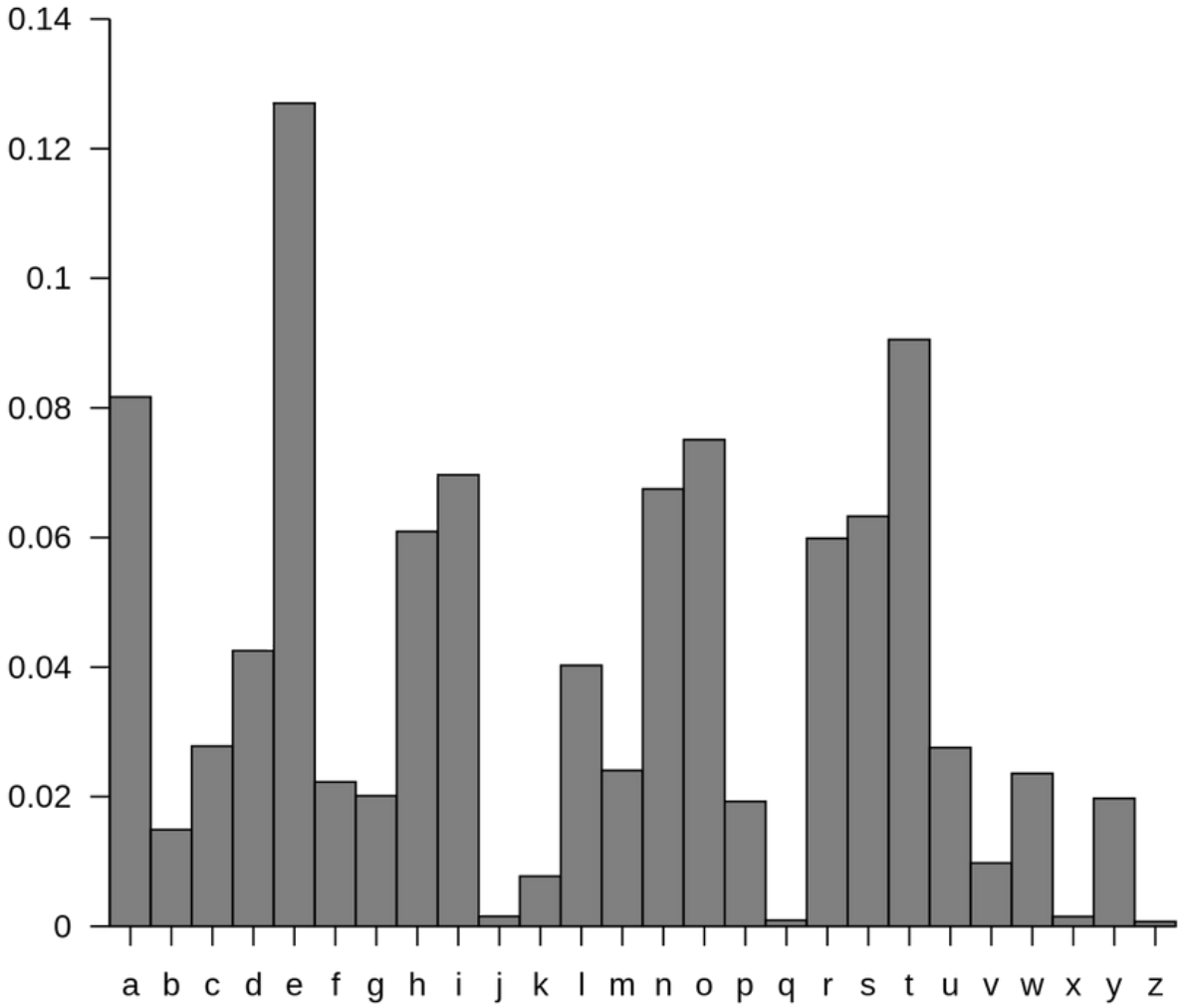
Source: Wikipedia

You can use two pieces of paper, or paper plates, attached in the middle by a brad to create your cipher wheel. **Figure out how to communicate changes in the wheel position mid-message for an added challenge. Hint: Use an agreed upon index letter.**

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FREQUENCY

Frequency analysis, developed by Arab mathematician Al-Kindi, and informed by the work of Al-Khalil, helps with deciphering. **How would you use this information?**

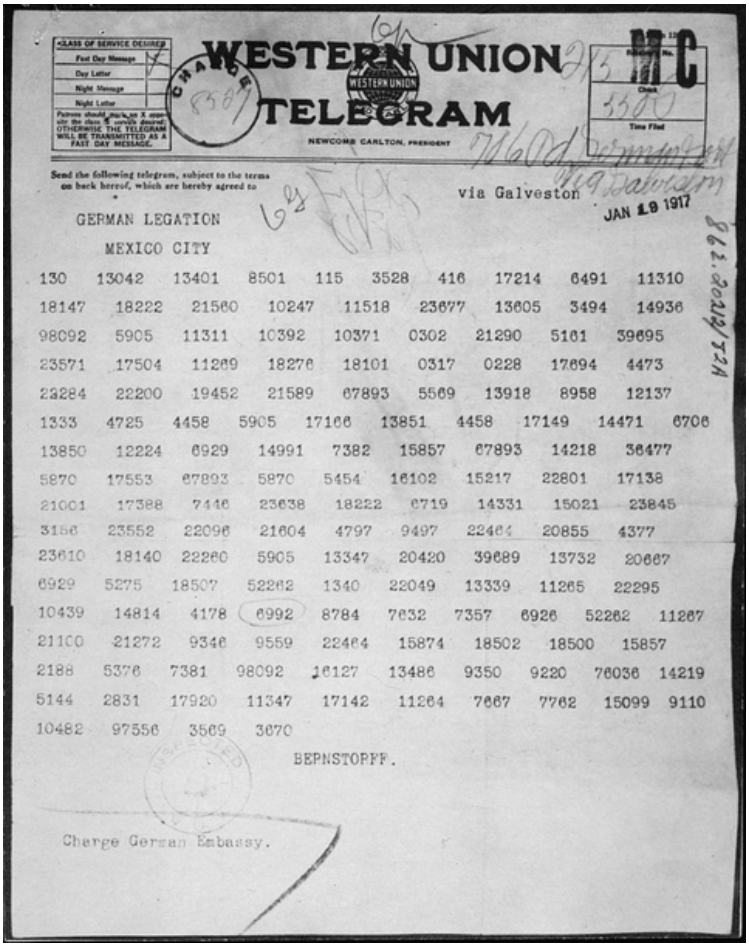


Source: Wikipedia

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ZIMMERMANN

Transport yourself to January 1917. An intercepted telegram led to America's entry into World War I. **Why?**



WESTERN UNION TELEGRAM

NEWCOMB CARLTON, PRESIDENT

Send the following telegram, subject to the terms on back hereof, which are hereby agreed to

via Galveston

JAN 19 1917

GERMAN LEGATION

MEXICO CITY

130 13042 13401 8501 115 3528 416 17214 6491 11310
18147 18222 21560 10247 11518 23677 13605 3494 14936
98092 5905 11311 10392 10371 0302 21290 5161 39695
23571 17504 11269 18276 18101 0317 0228 17694 4473
23284 22200 19452 21589 67893 5569 13918 8958 12137
1333 4725 4458 5905 17166 13851 4458 17149 14471 6706
13850 12224 6929 14991 7382 15857 67893 14218 36477
5870 17553 67893 5870 5454 16102 15217 22801 17138
21001 17388 7446 23638 18222 6719 14331 15021 23845
3156 23552 22096 21604 4797 9497 22464 20855 4377
23610 18140 22260 5905 13347 20420 39689 13732 20667
6929 5275 18507 52262 1340 22049 13339 11265 22295
10439 14814 4178 6992 8784 7632 7357 6926 52262 11267
21100 21272 9346 9559 22464 15874 18502 18500 15857
2188 5376 7381 98092 16127 13486 9350 9220 76036 14219
5144 2831 17920 11347 17142 11264 7667 7762 15099 9110
10482 97556 3569 3070

BERNSTORFF.

Charge German Embassy.

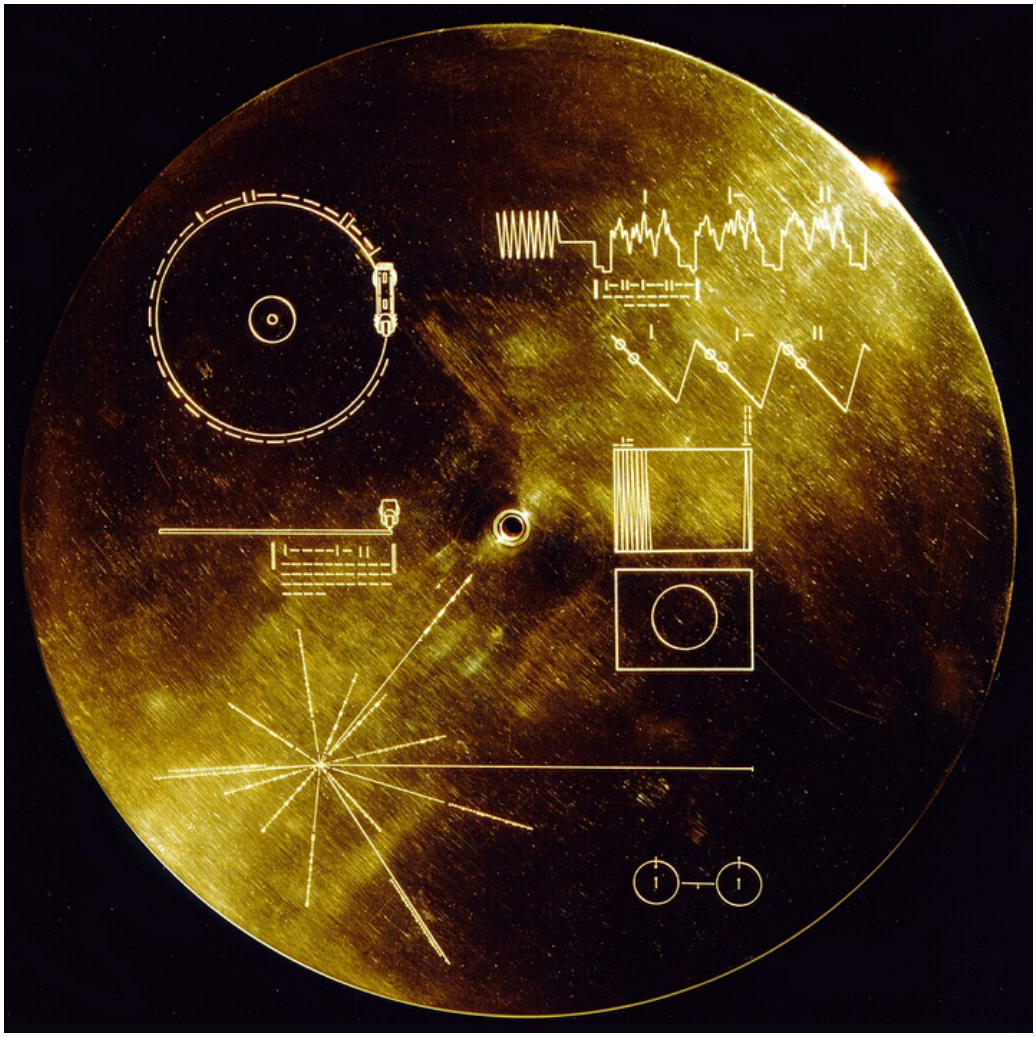
813.2024/124

- Zimmermann Telegram (Wikipedia)
- Zimmermann Telegram (National Archives)
- Zimmermann Telegram (National Security Agency)

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VOYAGER

The *Voyager* probes both contain records containing sounds and images from Earth. **Can you decipher the cover diagram?**



- The Voyager Golden Record (Wikipedia)

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CHALLENGE 1

Develop a cipher of your own, and test how long it takes you to code the messages below. If you have another person to work with, you can have more fun!

Develop a cipher together, and take turns passing coded messages, with one of you serving as Headquarters, and the other person serving as the Special Agent. All the messages that you pass must use the cipher that you developed! See if you can complete the entire sequence in under ten minutes, and if you have multiple teams at work, the fastest team captures the spy!

Remember all messages must be fully encrypted to be passed, and decrypted before a response may be sen b

Headquarters: What is the code word?
Special Agent: Butterfly Wings
Headquarters: Where is the spy?
Special Agent: 123 Spybuster Street
Headquarters: Police en route.
Special Agent: We captured the spy!

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CHALLENGE 2

To extraterrestrial life, our language would be a code that would need to be cracked before they could communicate with us.

You are tasked by the National Aeronautics and Space Administration (NASA) with developing a plaque for the next deep space mission: a series of satellites sent out with the express purpose of being found by intelligent life.

Design the plaque using a single piece of paper.

If you are working with others, share your plaque *without any instructions*. See if others can interpret your plaque correctly, and whether you can interpret theirs.

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UNBROKEN

Try these unbroken ciphers for the ultimate challenge:

- The Dorabella Cipher
- The D'Agapeyeff Cipher
- Pigeon NURP 40 TW 194
- Kryptos
- The Voynich Manuscript
- The Beale Ciphers
- The Shugborough Inscription

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